

Relationship between canopy disturbance history and current sapling density of *Fagus grandifolia* and *Acer saccharum* in a northern hardwood landscape

Philippe Nolet, Daniel Bouffard, Frédéric Doyon, and Sylvain Delagrange

Abstract: To shed light on the currently increasing proportion of *Fagus grandifolia* Ehrh. saplings in the tolerant hardwood forests of Quebec, we studied 48 *Acer saccharum* Marsh. dominated stands with contrasting histories of canopy disturbance: old commercial clear-cutting (CC), old fire (F), and either one or two partial cuts (1PC and 2PC). Our results indicated that higher densities of both *F. grandifolia* and *A. saccharum* saplings were associated with partial cutting histories (1PC and 2PC) than with severe canopy disturbance (CC and F). The density of *F. grandifolia* saplings was not related to any soil or stand characteristics in stands with a history of severe canopy disturbance. However, in stands with a history of partial canopy disturbances, the relative density of *F. grandifolia* saplings as compared with *A. saccharum* was related to soil C/N ratio and the presence of *F. grandifolia* overstory trees, whereas the absolute density showed a negative relationship with stand basal area. Therefore, it appears that partial canopy disturbances favored the regeneration of *F. grandifolia* relative to *A. saccharum*, whereas severe canopy disturbances may have provided an advantage to *A. saccharum*. We suggest that the presence of a light threshold can explain this shift in sapling performance between these two species.

Résumé : Pour tenter d'élucider l'augmentation de la proportion de gaule de *Fagus grandifolia* Ehrh. observée dans les forêts de feuillus tolérants à l'ombre du Québec, nous avons étudié 48 peuplements dominés par *Acer saccharum* Marsh. Ces peuplements se distinguent par leur historique de perturbation de la canopée : vieille coupe à blanc commerciale (CC), vieux feu (F) et une ou deux coupes partielles (1CP et 2CP). Nos résultats indiquent que les fortes densités de gaules de *F. grandifolia* et d'*A. saccharum* étaient davantage associées à un historique de coupe partielle (1CP et 2CP) qu'à des perturbations plus sévères de la canopée (CC et F). La densité des gaules de *F. grandifolia* n'était reliée à aucune caractéristique du sol ou du peuplement dans les peuplements ayant un historique de perturbation sévère de la canopée. Cependant, dans les peuplements ayant un historique de coupe partielle, la densité relative des gaules de *F. grandifolia* par rapport à celles d'*A. saccharum* était reliée au rapport C/N et à la présence de tiges de *F. grandifolia* dans l'étage dominant, alors que la densité absolue était négativement reliée à la surface terrière du peuplement. Il semble donc que des perturbations partielles de la canopée favorisent la régénération de *F. grandifolia* par rapport à celle d'*A. saccharum*, alors que les perturbations sévères de la canopée procureraient un avantage à *A. saccharum*. Nous croyons qu'il existe un seuil de luminosité pouvant expliquer les différences de densité des gaules de ces deux espèces.

[Traduit par la Rédaction]

Introduction

The composition of temperate deciduous forests in north-eastern North America underwent major changes over the last two centuries (Whitney 1987; Foster 1992; Nolet et al. 2001; Brisson and Bouchard 2003). These changes, as reported in historical forest reconstruction studies, include increases in sugar maple (*Acer saccharum* Marsh.) (Siccama 1971; Whitney 1987; Jackson et al. 2000; Nolet et al. 2001; Bouffard et al. 2003) and decreases in American beech (*Fagus grandifolia* Ehrh.) (Cogbill 2000). In contrast, several recent surveys showed an increase in *F. grandifolia* in

A. saccharum stands. For instance, Doyon et al. (2003) estimated that the relative dominance of *F. grandifolia* is expected to increase in more than 600 000 ha of forest in southern Quebec. Similarly, Duchesne et al. (2005) reported that sapling density of *F. grandifolia* doubled over a 10 year period in several deciduous forests of Quebec, whereas *A. saccharum* decreased by nearly 20% over the same period. From an economic perspective, the relative increase in *F. grandifolia* density may be problematic because its value is low compared with *A. saccharum*.

Several factors are thought to contribute to the increase of *F. grandifolia* over *A. saccharum*. For example, its high shade tolerance affords *F. grandifolia* higher relative survival under a closed canopy (Forcier 1975; Woods 1979) or after low-severity canopy disturbances (Poulson and Platt 1996). Moreover, white tailed deer (*Odocoileus virginianus* Zimmermann, 1780) preferentially browse other hardwood species over *F. grandifolia* and have affected regeneration success in some regions (Sage et al. 2003; Tremblay et al. 2007). Also, *F. grandifolia* is better adapted than *A. saccha-*

Received 26 February 2007. Resubmitted 30 July 2007.
Accepted 6 August 2007. Published on the NRC Research Press
Web site at cjfr.nrc.ca on 5 February 2008.

P. Nolet,¹ D. Bouffard, F. Doyon, and S. Delagrange. Institut
Québécois d'Aménagement de la forêt feuillue (IQAFF), 58
Principale, Ripon, QC J0V 1V0, Canada.

¹Corresponding author (e-mail: pnolet@iqaff.qc.ca).

rum to soils of low fertility (Kobe et al. 2002) or high acidity (Bouliane 1962). In fact, low calcium or high aluminum concentrations in soils have little effect on diameter growth of *F. grandifolia* saplings, whereas low soil calcium content can significantly increase mortality of *A. saccharum* (Kobe et al. 2002; Hane et al. 2003). Furthermore, *F. grandifolia* can reproduce vegetatively from root sprouts (Ward 1961; Jones and Raynal 1987), an adaptation that is associated with improved growth and survival under closed canopies relative to seedlings originating from sexual reproduction (Gardner and Mangel 1999). Finally, the crowns of established *F. grandifolia* tend to spread laterally (Beaudet and Messier 1998) and become wider, denser, and deeper than those of *A. saccharum* (Canham et al. 1994; Beaudet et al. 1999), thereby casting a deep shade (Horn 1971; Canham et al. 1994) that can provide a relative advantage to *F. grandifolia* regeneration.

Several studies have examined the coexistence and relative success of *F. grandifolia* and *A. saccharum* as a function of small to moderate canopy disturbances such as tree fall gaps or partial cuttings (Poulson and Platt 1996; Krasny and DiGregorio 2001; Arie and Lechowicz 2002). However, these studies have rarely been analyzed in light of applied research examining stand-level effects of different harvesting intensities on hardwood regeneration (e.g., Curtis and Rushmore 1958; Leak 1988; Houston 2001). Such an analysis is important because, although gap-phase disturbances are recognized as the principal canopy disturbance of tolerant hardwood forests (Runkle 1985; Canham 1989; Seymour et al. 2002), numerous stands in these forests have originated from large-scale disturbances such as fires or clearcuts (Nolet et al. 2001). The presence of such stands behooves an examination of how disturbances larger and more severe than tree fall gaps can affect the relative advantage of *F. grandifolia* over *A. saccharum*. Therefore, our study focused on the possible footprint left by a wide range of canopy disturbance that occurred during the 20th century on the current density of *F. grandifolia* saplings in *A. saccharum* dominated stands. Our objectives were (i) to evaluate whether the current densities of *F. grandifolia* and *A. saccharum* saplings varied as a function of contrasting histories of canopy disturbance and (ii) to examine whether this variation in sapling density was associated with current stand and soil characteristics.

Methods

Study area

The principal study area was in the Papineau-Labelle Wildlife Reserve (46°28'N, 75°35'W), 94 km north of the city of Gatineau, Quebec. This 144 km² area includes a large part of the Lac Poisson Blanc landscape unit (Robitaille and Saucier 1998), which is within the sugar maple – yellow birch bioclimatic zone (Saucier et al. 1998). Soils are typically thin glacial tills containing metamorphic rocks, generally topped by sandy dystic brunisols (Agriculture Canada 1990). The landscape has numerous hills with maximum and mean elevations of 450 and 300 m, respectively. Mean annual temperature and precipitation are 3.7 °C and 1100 mm (of which 250 mm is snow). On average, the area experiences 2716 degree-days/year above 0 °C (Environment Canada 2007). Stand composition is typically a mix of *A. saccharum* or yellow birch (*Betula*

alleghaniensis Britton), American basswood (*Tilia americana* L.), ironwood (*Ostrya virginiana* (Mill.) K. Koch), *F. grandifolia*, and balsam fir (*Abies balsamea* (L.) Mill.) and depends on slope and drainage.

Disturbance history types

Based on the disturbance histories reconstructed by Nolet et al. (2001), we selected 48 *A. saccharum* dominated stands with four different histories of canopy disturbance (i.e., 12 stands per disturbance history type (DHT)): commercial clearcuts implemented approximately 50 years ago (CC); stands originating after a fire that burned approximately 80 years ago (F); stands with two partial cuts conducted during the previous 50 years (2PC); and stands with one partial cut implemented during the last 25 years (1PC). In the CC stands, >75% of forest cover was removed between 1940 and 1960. The burned stands (F) originated from a single large and severe fire between 1920 and 1925 (most likely in 1923). According to the federal government in 1930 (MacLaren 1956. Forest management plan revision of the Lievre River timber limits. Decade 1956–1966.), the fire was “severe” for 75% of the hardwood cover type that burned. The 2PC stands were harvested between 1956 and 1971 with a diameter-limit cut that removed all trees ≥30 cm, followed by another partial cut between 1983 and 1995. The 1PC stands experienced one partial cut, between 1983 and 1995.

For each selected stand, we confirmed the history of past disturbance using management plans, historical maps, soil charcoal surveys, and dendrochronological analyses of dominant trees. In their reconstruction of disturbance history, Nolet et al. (2001) evaluated disturbance severity from aerial photographs using two categories: stands with <25% retained forest cover and stands with 25%–75% retained forest cover. The CC and F stands were in the first category while the 1PC and 2PC stands were in the latter. Although we did not more precisely quantify severity of canopy opening in this study, percent cover after the CC and F was likely closer to 0% than to 25% since we avoided stands in these DHTs that were stocked with trees having a diameter at breast height (DBH; 1.3 m) > 50 cm. That said, it remains possible that remnant seed trees within or adjacent to the CC and F stands influenced the development of the sampled stands. According to historical maps and photographs, and corroborated visually, 1PC stands were not harvested between 1915 and 1985. Although the severity and character of partial cuts implemented between the mid-1980s and mid-1990s is unknown, it is likely these cuts were similar to the treatments implemented in the first pass of the 2PC stands. During the early 1990s, policy changes mandated a shift from diameter-limit cutting to selection cutting in the public hardwood forests of Quebec. This shift occurred gradually, and the preliminary rules for selection cutting were similar to those of diameter-limit cutting, with the exception that harvest be limited to 35% of the initial stand basal area. Between 1956 and 1995, all species except eastern hemlock (*Tsuga canadensis* (L.) Carr.) were considered merchantable, irrespective of timber grade; therefore, we assumed no species were preferentially targeted. We ensured the selected stands were stocked with *F. grandifolia* seed trees, thus avoiding stands with low sapling density due to the absence of a source of *F. grandifolia* propagules.

As a comparison, we also analyzed sapling density in the Gatineau Experimental Forest, located about 40 km north-east of Gatineau (45°55'N, 76°05'W). Climate, soils, landscape, and forest composition are similar to the principal study area, but the number of degree-days is higher (2897; Environment Canada 2007). This concurrent sampling addressed the short-term effect of true clear-cutting (complete overstory removal) on *F. grandifolia* sapling density in stands with similar initial canopy composition. In this case, 30–90 m wide strips were clearcut in 1983 and 1985, leaving 60–180 m wide uncut strips of forest in between. Twelve matched strips (S) and between-strip areas (BS) were randomly sampled to compare differences in *F. grandifolia* sapling density between recently disturbed and undisturbed portions in these stands.

Field measurements

Within each selected stand in the principal study area, we established three sampling points at the corners of an equilateral triangle covering an area of approximately 0.2 ha (Fig. 1A). In the strip clearcuts of the Gatineau Experimental Forest, the three sampling points were distributed at 20 m intervals along transects in the cut and uncut areas (Fig. 1B). At each sampling point in both study areas, we recorded DBH and species for each tree ≥ 9.1 cm DBH, as determined using a factor 2 prism (metric). The abundance of *F. grandifolia* and *A. saccharum* saplings (1–9 cm DBH) was recorded on five 1.8 m radius plots (10 m²) (Figs. 1A and 1B). The first of these plots was located at the prism point centre, and the other four were established at 7 m in each of the four cardinal directions (Fig. 1). For each stand, six soil samples (two per point) were extracted from the B horizon and later aggregated to estimate average soil conditions for the stand ($n = 12$ composite samples for each DHT, for a total of 72 sampled stands).

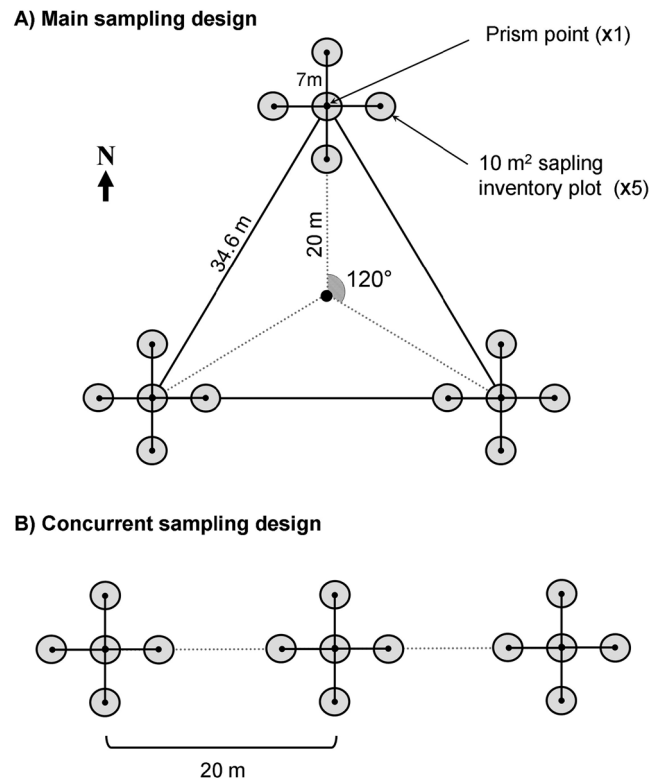
Analyses of soil chemistry

The soil samples were dried and sieved with a 2 mm lattice after measuring their C content by combustion using a carbon analyzer (Leco CR-412; Leco Corp., St. Joseph, Mich.). Soil N content was assessed using Kjeldahl digestion (Kjeltec Tecator 1030; Foss Corp., Brampton, Ont.) coupled with a spectrophotometric dosage with continuous flow (Quikchem 8000; Lachat Corp., Milwaukee, Wis.). From these measurements, soil C/N ratio was calculated. Exchangeable cations (K, Ca, Mg, Mn, Al, Fe, and S) were titrated by plasma atomic emission spectrometry (ICAP 9000, Thermo Instruments, Franklin, Mass.) in a soil solution obtained from dilution in a NH₄Cl solution (1 mol/L) for 12 h. Cation exchange capacity (CEC) was then computed as the sum of all measured exchangeable cations. Finally, soil pH was measured from a filtered soil solution using a pH meter equipped with a Ross Sure-Flow electrode (Orion 8172 BN; Orion Research, Beverly, Mass.).

Data analyses

The basal area of *A. saccharum* (BA_{AS}), *F. grandifolia* (BA_{FG}), other deciduous species (BA_{Deciduous}), and all conifer species (BA_{Conifers}) as well as for the total stand (BA_{Stand}) were calculated from the prism point measurements. Stand sapling density (SD_{Stand}), *F. grandifolia* sapling density

Fig. 1. Schematic representation of the sampling designs in the two study sites.



(SD_{FG}), and the relative density of *F. grandifolia* saplings compared with *A. saccharum* saplings (RD_{FG}) were then determined based on measurements from the 1.8 m plots, and averaged to estimate values for the stand.

We used one-way analysis of variance (ANOVA) to compare stand and soil characteristics among CC, F, IPC, and 2PC stands or between S and BS stands. An LSD test was conducted a posteriori to evaluate differences in stand traits among DHTs. Similarly, an ANOVA was used to compare density of saplings (SD_{FG} and RD_{FG}) among DHTs. Log transformations of dependent variables were applied when needed to satisfy normality and homoscedasticity assumptions.

Because preliminary analyses of stand and soil characteristics revealed little difference between the severe DHTs (CC and F) and between the partial DHTs (IPC and 2PC), we aggregated them into two disturbance history groups (DHGs). To investigate differences in sapling density of *F. grandifolia* (SD_{FG} and RD_{FG}) between the two DHGs, *t* tests were used because equality of variance was not achieved even after transformation. Further analyses of *F. grandifolia* sapling density were conducted separately for each DHG. To elucidate the relationships between density of *F. grandifolia* saplings and various soil and stand characteristics in each DHG, we conducted a Spearman's rank test to correlate SD_{FG} and RD_{FG} with BA_{Stand}, BA_{FG}, CEC, pH, C/N ratio, and content of Ca, K, Mg, and Al. Finally, predictive models for *F. grandifolia* sapling density were explored using multiple linear regressions for each DHG. Nine independent variables were analyzed for their relationships with SD_{FG} and RD_{FG} based on the following model forms:

Table 1. Characteristics of stand structure and stand composition as a function of disturbance history type (DHT).

Stand characteristic*								
DHT†	Age (years)	BA _{Stand} (m ² ·ha ⁻¹)	DBH _{Mean} (cm)	SD _{Stand} (trees·ha ⁻¹)	BA _{AS} (m ² ·ha ⁻¹)	BA _{FG} (m ² ·ha ⁻¹)	BA _{Deciduous} (m ² ·ha ⁻¹)	BA _{Conifers} (m ² ·ha ⁻¹)
CC	50–70	27.67 (1.41)b	21.42 (0.41)a	544 (120)a	21.50 (1.69)b	3.11 (0.48)a	2.78 (1.31)a	0.28 (0.13)a
F	70–80	26.33 (1.21)b	21.03 (0.76)a	3050 (311)b	13.72 (2.04)a	3.22 (0.84)a	8.94 (1.28)b	0.45 (0.24)a
2PC	Uneven aged	21.61 (1.26)a	23.84 (0.88)b	3400 (537)b	14.55 (1.51)a	4.17 (0.92)a	2.06 (0.50)a	0.83 (0.39)a
1PC	Uneven aged	22.06 (1.11)a	26.69 (1.30)c	6056 (745)c	13.33 (1.26)a	3.61 (1.06)a	3.56 (1.07)a	1.56 (0.63)a

Note: Values are means with SEs given in parentheses except for stand ages ($n = 12$ for each disturbance history type). Values within a column with different letters are significantly different at $\alpha = 0.05$.

*BA_{Stand}, stand basal area; DBH_{Mean}, mean stand diameter at breast height; SD_{Stand}, stand sapling density; BA_{AS}, basal area of *A. saccharum*; BA_{FG}, basal area of *F. grandifolia*; BA_{Deciduous}, basal area of other deciduous species; BA_{Conifers}, basal area of conifer species.

†CC, clearcut; F, fire; 2PC, two partial cuttings; 1PC, one partial cutting.

$$[1] \quad SD_{FG} = b_0 + b_1 (BA_{Stand}) + b_2 (BA_{FG}) \\ + b_3 (\text{soil C/N}) + b_4 (K) + b_5 (Ca) \\ + b_6 (Mg) + b_7 (Al) + b_8 (CEC) \\ + b_9 (pH)$$

$$[2] \quad RD_{FG} = b_0 + b_1 (BA_{Stand}) + b_2 (BA_{FG}) \\ + b_3 (\text{soil C/N}) + b_4 (K) + b_5 (Ca) \\ + b_6 (Mg) + b_7 (Al) + b_8 (CEC) \\ + b_9 (pH)$$

where b_0 – b_9 are regression coefficients.

Prior to the multiple regression analyses, log transformations were performed on the independent variables where needed to obtain relationship linearity. A stepwise procedure was used to incrementally include independent variables that significantly increased the proportion of explained variance, thereby providing the most parsimonious final model that explained the most variation in our dependent variable. All statistical analyses were performed using NCSS 2001 software (Hintze 2002). Differences were considered significant at $\alpha = 0.05$.

Results

Stand characteristics

Disturbance history significantly affected several stand characteristics. The BA_{Stand} was significantly higher in CC and F stands than in 1PC and 2PC stands (Fig. 1, Tables 1 and 2). We observed significant differences in the arithmetic mean DBH for the stand (DBH_{Mean}) and SD_{Stand} among DHTs (Fig. 1, Tables 1 and 2). DBH_{Mean} and SD_{Stand} were greater in the partially cut stands than in CC stands, but no significant differences were observed between F and CC for the former variable, or between F and 2PC for the latter. Canopy composition was similar among DHTs, as estimated by BA_{Conifers} and BA_{FG} (Tables 1 and 2). However, BA_{AS} was significantly higher in CC stands than in the other DHTs, and BA_{Deciduous} was significantly higher in F than in CC or partially cut stands (Tables 1 and 2).

Soil properties

With one exception, no significant difference in soil properties was observed among DHTs. No significant differences were reported for soil C/N ratio among DHTs (Fig. 2A, Ta-

Table 2. Summary of F ratio and p values for ANOVAs of stand characteristics and soil characteristics (soil C/N ratio; soil pH; exchangeable capacity of potassium (K), calcium (Ca), and magnesium (Mg); and total cation exchangeable capacity (CEC)) as a function of disturbance history type (DHT).

Source of variation	ANOVA results	
	F	p
Stand characteristics		
BA _{Stand}	5.89	0.0018
DBH _{Mean}	8.48	0.0001
SD _{Stand}	21.31	<0.0001
BA _{FG}	0.31	0.1057
BA _{AS}	5.71	0.0022
BA _{Deciduous}	6.44	0.001
BA _{Conifers}	0.66	0.5837
Soil (B horizon) characteristics		
Soil C/N	0.96	0.4209
K	5.38	0.0030
Ca	1.21	0.3165
Mg	1.50	0.2287
CEC	0.84	0.4774
pH	1.68	0.1860

Note: See Table 1 for abbreviations.

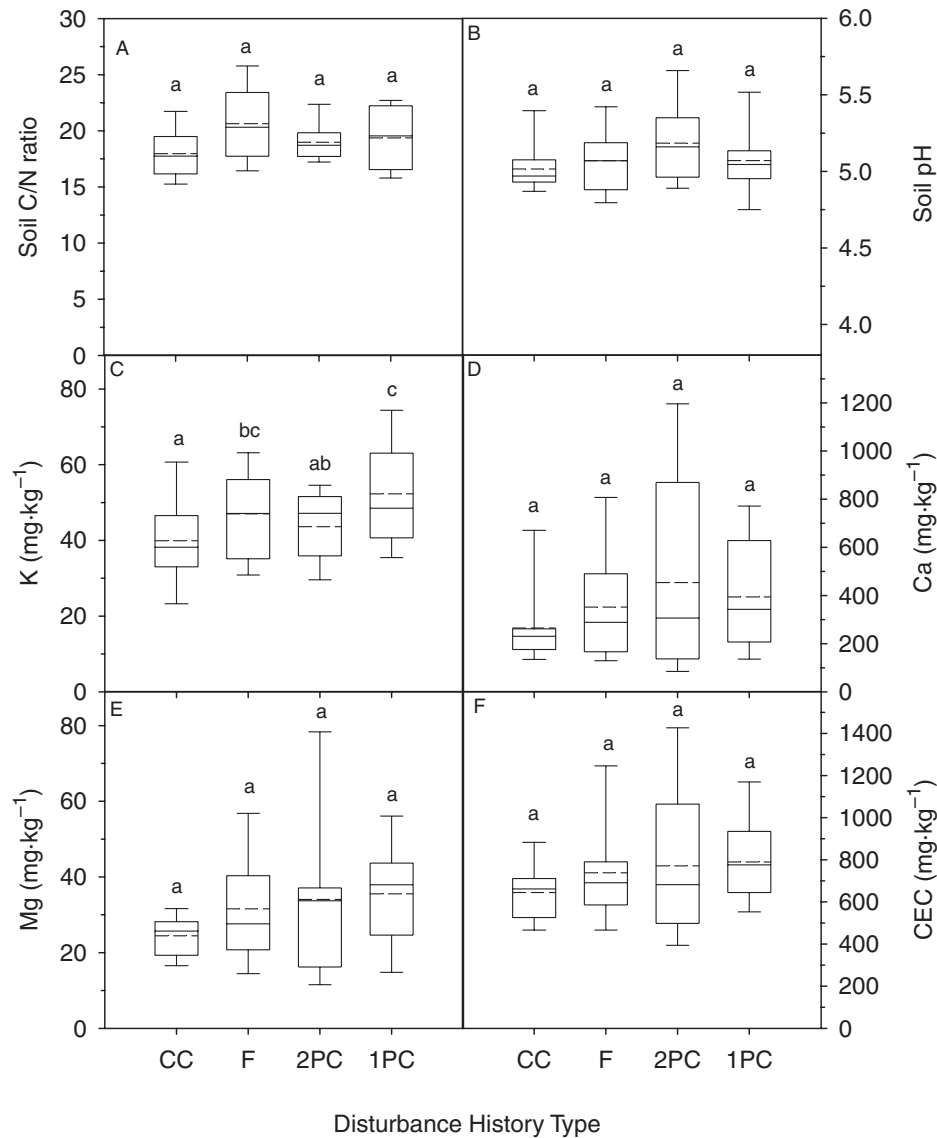
ble 2). Despite slightly higher mean values for soil pH, Ca, Mg, and CEC in 1PC and 2PC when compared with F and CC, differences were not significant (Figs. 2B, 2D, 2E, and 2F, Table 2). Only soil K content differed significantly among DHTs (Fig. 2C, Table 2).

Density of *F. grandifolia* saplings

The SD_{FG} was significantly higher in 1PC and 2PC DHTs (Fig. 3A) than in CC DHT, whereas the F DHT was not different from any others. Variation of SD_{FG} was also much greater in 1PC and 2PC DHTs than in F or CC DHTs (Fig. 3A and 3B). Furthermore, RD_{FG} did not differ significantly among DHTs (Figs. 3C and 3D). Disturbance severity affected the absolute, but not relative, density of both species.

Stands with a high SD_{FG} (i.e., >1500 saplings/ha) corresponded to 0%, 16.7%, 41.7%, and 50% of the CC, F, 1PC, and 2PC DHTs, respectively. On the other hand, at the Gati-neau Experimental Forest, we observed no difference in

Fig. 2. Box plots for (A) soil C/N ratio, (B) soil pH, (C) soil K content, (D) soil Ca content, (E) soil Mg content, and (F) total cation exchangeable capacity (CEC) as a function of disturbance history type. Boxes are the 5th to 95th percentiles, solid lines in the boxes are the medians, and broken lines are the means. CC, clearcut; F, fire; 2PC, two partial cuttings; 1PC, one partial cutting. Letters indicate significant differences at $\alpha = 0.05$ in the ANOVA a posteriori LSD test.



SD_{FG} between the S and BS plots. However, RD_{FG} is significantly lower in S plots than in BS plots ($p = 0.0005$, Fig. 4).

Relationship between site characteristics and *F. grandifolia* sapling density

As mentioned above, stand and sapling density analyses prompted the use of only two broad groups (DHG) of DHTs (severe, CC and F, vs. partial, 2PC and 1PC). The lack of significant differences in soil properties among stands gave credence to this grouping and the implementation of separate analyses for the two groups. Spearman's correlation analyses between SD_{FG} and RD_{FG} and stand and soil characteristics (Table 3) showed two contrasting patterns of *F. grandifolia* sapling density according to DHG. For the severe DHG, SD_{FG} and RD_{FG} were not significantly correlated with any stand and soil characteristics (Table 3). However, for the partial DHG, SD_{FG} and RD_{FG} were strongly negatively correlated to BA_{Stand}

and strongly positively correlated to BA_{FG} . Moreover, for the partial DHG, strong positive correlations were found between soil C/N ratio and SD_{FG} and RD_{FG} , whereas negative correlations were observed between soil Mg content and SD_{FG} and RD_{FG} . Furthermore, SD_{FG} also showed negative correlations with soil CEC and Ca content. These correlations were not significant for RD_{FG} (Table 3).

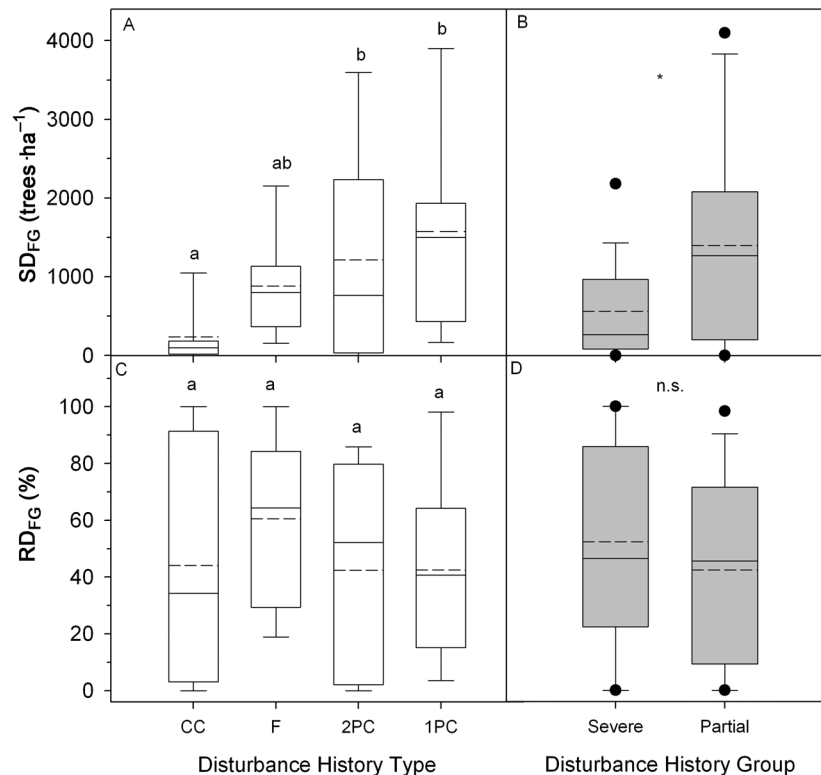
For the severe DHG, no multiple linear regressions significantly explained the variation of SD_{FG} and RD_{FG} . However, for the partial DHG, SD_{FG} was positively associated with soil C/N ratio and negatively with BA_{Stand} , while RD_{FG} was positively related to soil C/N ratio and BA_{FG} (Table 4).

Discussion

Disturbance history and density of *F. grandifolia* saplings

We examined the density of *F. grandifolia* sapling in var-

Fig. 3. Box plots for (A and B) relative density (RD_{FG}) and (C and D) sapling density (SD_{FG}) of *F. grandifolia* saplings as a function of disturbance history type (CC, clearcut; F, fire; 2PC, two partial cuttings; 1PC, one partial cutting) or disturbance history group (CC and F, severe; 1PC and 2PC, partial). Boxes are the 5th to 95th percentiles, solid lines in the boxes are the medians, and broken lines are the means. Letters indicate significant differences between DHT for RD_{FG} and SD_{FG} at $\alpha = 0.05$ in the ANOVA a posteriori LSD test (A and C). In the t test (B and D), the asterisk indicates a significant difference at $\alpha = 0.05$. ns, not significant ($p > 0.05$).



ious stages of stand development: saplings (S plots); poles (F and CC plots) and small sawtimber (1PC, 2PC, and BS plots). At present, the relative density of *F. grandifolia* saplings was lower in 20-year-old S plots than in adjacent uncut BS plots of the same stand (Fig. 4). Based on the canopy composition of CC and F stands dominated by *A. saccharum* (Table 1), it is likely the high current relative density of *A. saccharum* in clearcut strips will lead to *A. saccharum* dominated stands. Other authors found open conditions favour *A. saccharum* over *F. grandifolia* (Bicknell 1982; Nyland et al. 2004). Unfortunately, we cannot ascertain whether this *A. saccharum* dominance is due to a competitive advantage of *A. saccharum* under high light conditions or to the conditions present immediately after disturbance, because we have no information regarding advance regeneration, the relative abundance of residual *F. grandifolia* and other species immediately after harvest, or other information regarding initial conditions of our study stands. Stands with severe disturbance history (CC and F) had a low current sapling density of both species. These low densities are probably associated with the high basal area of the stands (Table 1) that reduces the amount of light reaching the sapling layer. Furthermore, stands in the small sawtimber stage that experienced a partial canopy disturbance history (1PC and 2PC) had the highest density of *F. grandifolia* saplings (Fig. 3). This result suggests *F. grandifolia* may be favored by partial canopy disturbances and corroborates several other studies (e.g., Krasny and DiGregorio 2001; Rhoads et al. 2002; Angers et al. 2005).

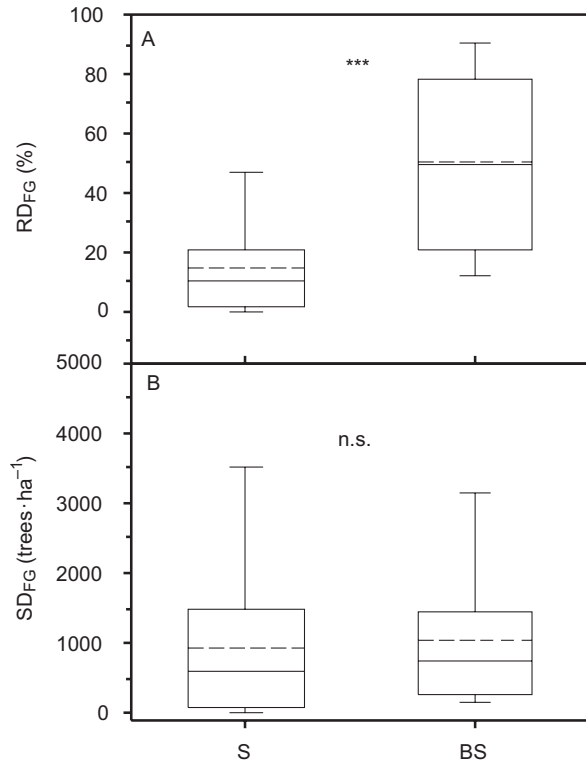
Our study shows that, at least in our study area, different canopy disturbance histories during the 20th century influenced the current dynamics of *F. grandifolia* and *A. saccharum* saplings in *A. saccharum* dominated stands (Fig. 3). It is likely that other factors, such as preferential browsing by white-tailed deer on *A. saccharum* (Sage et al. 2003), *A. saccharum* decline (Horsley et al. 2002; Duchesne et al. 2005), or atmospheric depositions (Ouimet and Camiré 1995; Long et al. 1997), may have interacted with canopy disturbance to influence the dynamics of our two focal species. Although beech bark disease (*Nectria coccinea* Desm.) has also been reported as a contributing factor to the increase of *F. grandifolia* saplings beneath tolerant hardwood stands—principally by enhancing beech root sprouts (Le Guerrier et al. 2003)—this disease has low incidence north of the St. Lawrence River in Quebec.

Current site characteristics and *F. grandifolia* sapling density

Two contrasting patterns were elucidated regarding the relationship between current density of *F. grandifolia* and current site characteristics as a function of history of canopy disturbances:

1. In old stands that experienced severe canopy disturbances (i.e., F and CC stand), current *F. grandifolia* sapling density was not associated with any current stand or soil characteristics. As discussed above, these stands now have a high basal area that reduces the light availability

Fig. 4. Box plots for (A) relative density (RD_{FG}) and (B) sapling density (SD_{FG}) of *F. grandifolia* saplings in the strip (S) – between strip (BS) system. Boxes are the 5th to 95th percentiles, solid lines in the boxes are the medians, and broken lines are the means. Asterisks show statistical significance: ***, $p \leq 0.001$; ns, not significant ($p > 0.05$).



necessary for development of the sapling layer. Under such light limitations, it is possible that the effects of other factors become negligible, especially soil characteristics (Coomes and Grubb 2000; Ricard et al. 2003).

2. In stands with a history of partial canopy disturbances, the current density of *F. grandifolia* saplings was correlated with both soil (C/N, CEC, Mg, and Ca) and stand characteristics (BA_{Stand} and BA_{FG}). Among soil variables, C/N was the most important factor explaining the observed variation in SD_{FG} and RD_{FG} (Table 4). The positive relationship between C/N ratio and SD_{FG} was expected, because *F. grandifolia* has a competitive advantage over *A. saccharum* on poor sites (e.g., Moore et al. 2000; Bigelow and Canham 2002; Duchesne et al. 2002; Kobe et al. 2002; Hane et al. 2003). It is noteworthy that Ca and Mg, which are recognized as important influences on the regeneration dynamics of *F. grandifolia* and *A. saccharum*, do not appear as significant variables in the multiple regression models, because they did not explain enough additional variation of the dependent variable after C/N ratio was included.

In terms of stand factors, variation in RD_{FG} in stands with a history of partial canopy disturbances was positively related to current stand BA_{FG} . In addition to the fact that a higher proportion of BA_{FG} in the canopy should increase the seed sources of this species, at least two mechanisms have been identified in the literature linking the success of *F. grandifolia* regeneration beneath its own canopy. First, a

Table 3. Results of Spearman's rank correlation test for sapling density (SD_{FG}) and relative density (RD_{FG}) as a function of stand basal area (BA_{Stand}), basal area of *F. grandifolia* (BA_{FG}), soil C/N ratio; soil pH; exchangeable capacity of soil potassium (K), calcium (Ca), magnesium (Mg), and aluminium (Al); and total soil cation exchangeable capacity (CEC).

	Density of <i>F. grandifolia</i> saplings	
	SD_{FG}	RD_{FG}
Severe disturbance history group (CC and F)		
Stand characteristics		
BA_{Stand}	-0.25	-0.18
BA_{FG}	0.05	0.04
Soil (B horizon) characteristics		
Soil C/N	0.13	0.13
K	0.23	-0.11
Ca	-0.03	-0.13
Mg	0.01	-0.12
Al	-0.16	-0.12
CEC	-0.07	-0.22
pH	0.27	0.11
Partial disturbance history group (1PC and 2PC)		
Stand characteristics		
BA_{Stand}	-0.68***	-0.48*
BA_{FG}	0.58**	0.67***
Soil (B horizon) characteristics		
Soil C/N	0.63**	0.69***
K	-0.11	-0.10
Ca	-0.58**	-0.38
Mg	-0.61**	-0.44*
Al	0.24	0.21
CEC	-0.53**	-0.31
pH	-0.21	-0.17

Note: Correlation analyses were performed after grouping clearcut (CC) and fire (F) into a severe disturbance history group and the two partial cutting types (2PC and 1PC) into a partial disturbance history group ($n = 24$ per group). Asterisks indicate statistical significance: *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$.

high proportion of *F. grandifolia* in the canopy reduces light transmission to the understory (Canham et al. 1994; Beaudet et al. 2002), which is more detrimental for *A. saccharum* than for *F. grandifolia* growth (Finzi and Canham 2000; Ricard et al. 2003). Second, high BA_{FG} may increase the quantity of *F. grandifolia* litter in the forest floor, generating conditions more favorable for *F. grandifolia* than *A. saccharum* seedlings (Hane et al. 2003). However, our data do not allow us to evaluate how these mechanisms may have interacted. Finally, SD_{FG} and RD_{FG} are negatively correlated with BA_{Stand} . This result suggests that high light levels favor *F. grandifolia* saplings over *A. saccharum* saplings in stands with partial canopy disturbance history. This result is surprising because it contradicts our results from the strip cuts that suggest that high light levels favor *A. saccharum* over *F. grandifolia* saplings.

Implications for management and research

Poulson and Platt (1996) suggested large and frequent canopy gaps should be more favorable to *A. saccharum* than to *F. grandifolia* regeneration. Our findings are not com-

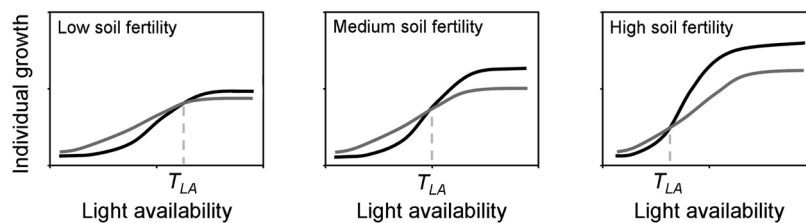
Table 4. Results of the predictive models for density density (SD_{FG}) and relative density (RD_{FG}) of *F. grandifolia* saplings as a function of disturbance history group (DHG).

DHG	Dependant variable	Predictive multiple linear model	Model parameters		
			R^2	Adjusted R^2	p
Severe	SD_{FG}	None	na	na	>0.05
	RD_{FG}	None	na	na	>0.05
Partial	SD_{FG}	$1386.46 + 205.68(C/N) - 179.23(BA_{Stand})$	0.57	0.53	<0.001
	RD_{FG}	$-94.71 + 6.28(C/N) + 4.60(BA_{FG})$	0.53	0.48	<0.001

Note: See Table 3 for definitions of the DHGs.

Fig. 5. Hypothesized effect of light availability on the relative performance of *F. grandifolia* (shaded line) and *A. saccharum* (solid line) saplings as a function of site quality. The light availability threshold (T_{LA}) corresponds to the intersection of *A. saccharum* and *F. grandifolia* lines.

Growth potential at the individual level



pletely consistent with this model. In fact, the most important result of this study is probably that a regime of partial canopy disturbances may favor the regeneration of *F. grandifolia* relative to *A. saccharum*, whereas much more severe canopy disturbances such as clearcuts or fires may provide an advantage to the regeneration of *A. saccharum* over *F. grandifolia*. Seemingly, this shift in performance between both species is related to the available light. We suggest that a critical light threshold exists and that it corresponds to the intersection of the growth curves of *A. saccharum* and *F. grandifolia*, as shown in Fig. 5A. Several authors have shown that *A. saccharum* grows better than *F. grandifolia* under high light (Bicknell 1982; Nyland et al. 2004), and grows relatively less at low light (Ricard et al. 2003; Leak 2004). Because the growth rates of these species vary differently in relation to soil chemical conditions (Long et al. 1997; Kobe et al. 2002), it is likely this critical light threshold also varies with soil chemical conditions (Fig. 5). Consequently, we hypothesize that larger openings will be necessary to favor *A. saccharum* over *F. grandifolia* on nutrient-limiting sites.

Other factors may significantly influence the outcome of severe or partial disturbances. First, when severe disturbances occur, the outcome may not systematically be *A. saccharum* dominated stands as observed in this study, because the abundance and size of advanced *A. saccharum* or *F. grandifolia* regeneration may significantly affect which species ultimately dominates. It is possible the development of an *A. saccharum* dominated stand after commercial clearcuts and fires was related to the absence of understory *F. grandifolia*, its destruction during the disturbance, or to the paucity of large *F. grandifolia* residual trees that can help initiate a new cohort. The importance of advanced regeneration was illustrated by studies that have shown that, when high levels of *F. grandifolia* saplings are found before harvest, clearcuts often lead to a dominance of *F. grandifolia* in the new co-

hort (Richards and Farnworth 1971; Kelty and Nyland 1981). Second, partial disturbances may not systematically favor an increase of *F. grandifolia*. The abundance of *F. grandifolia* seed sources and soil richness, as suggested by our data, in conjunction with advanced regeneration may influence the outcome of *A. saccharum* and *F. grandifolia* regeneration. Consequently, more research is needed to confirm and quantify the light threshold and to understand how it interacts with other factors, such as soil richness and understory conditions at the time of disturbance.

Acknowledgments

We gratefully acknowledge the critical contributions of the following people for the completion of this study: Régis Pouliot, Denis F. Doyon, Benoit Lafleur, and Mario Lajeunesse offered essential help collecting field data. We express our gratitude to Carol De Blois from the chemistry laboratory of the Direction de la recherche forestière (ministère des Ressources naturelles et de la Faune du Québec, MRNFQ) for the chemical analyses. We also appreciate the support and scientific advice provided by Louis Duchesne (MRNFQ) at the beginning of the project. We thank Stephen Yamasaki for help with statistical analyses, Dominique Gravel and Robin Duchesneau for enlightening discussions, and Ronnie Drever for English editing. We also thank four anonymous reviewers who significantly contributed to this paper. This study was supported by a grant from the Programme de Mise en Valeur des Ressources du Milieu Forestier through partnership between the MRNFQ and Lauzon Forest Industries Inc.

References

- Agriculture Canada. 1990. Pêdo-paysages du Canada: sud-ouest du Québec. Agriculture Canada, Unité de cartographie et des systèmes d'information, Centre de recherches sur les terres, Direction générale de la recherche, Montréal, Que.

- Angers, V.A., Messier, C., Beaudet, M., and Leduc, A. 2005. Comparing composition and structure in old-growth and harvested (selection and diameter-limit cuts) northern hardwood stands in Quebec. *For. Ecol. Manage.* **217**: 275–293. doi:10.1016/j.foreco.2005.06.008.
- Arii, K., and Lechowicz, M.J. 2002. The influence of overstory trees and abiotic factors on the sapling community in an old-growth *Fagus–Acer* forest. *Ecoscience*, **9**: 386–396.
- Beaudet, M., and Messier, C. 1998. Growth and morphological responses of yellow birch, sugar maple, and beech seedlings growing under a natural light gradient. *Can. J. For. Res.* **28**: 1007–1015. doi:10.1139/cjfr-28-7-1007.
- Beaudet, M., Messier, C., Paré, D., Brisson, J., and Bergeron, Y. 1999. Possible mechanisms of sugar maple regeneration failure and replacement by beech in the Boisé-des-Muir old-growth forest, Québec. *Ecoscience*, **6**: 264–271.
- Beaudet, M., Messier, M., and Canham, C.D. 2002. Predictions of understory light conditions in northern hardwood forests following parameterization, sensitivity analysis, and tests of the SORTIE light model. *For. Ecol. Manage.* **165**: 235–248. doi:10.1016/S0378-1127(01)00621-1.
- Bicknell, S.H. 1982. Development of canopy stratification during early succession in northern hardwoods. *For. Ecol. Manage.* **4**: 41–51. doi:10.1016/0378-1127(82)90027-5.
- Bigelow, S., and Canham, C.D. 2002. Community organization of tree species along soil gradients in a north-eastern USA forest. *J. Ecol.* **90**: 188–200. doi:10.1046/j.0022-0477.2001.00655.x.
- Bouffard, D., Doyon, F., and Forget, E. 2003. Historique et dynamisme écologique de la végétation forestière de la réserve faunique Rouge-Matawin de 1930 à nos jours. Institut québécois d'Aménagement de la Forêt feuillue, Ripon, Que.
- Bouliane, A. 1962. Étude de la transformation de l'érablière à sucre en hêtraie. Mémoire de Maîtrise, Faculté de Foresterie et de Géomatique, Université Laval, Québec, Que.
- Brisson, J., and Bouchard, A. 2003. In the past two centuries, human activities have caused major changes in the tree species composition of southern Quebec, Canada. *Ecoscience*, **10**: 236–246.
- Canham, C.D. 1989. Different responses to gaps among shade-tolerant tree species. *Ecology*, **70**: 548–550. doi:10.2307/1940200.
- Canham, C.D., Finzi, A.C., Pacala, S.W., and Burbank, D.H. 1994. Causes and consequences of resource heterogeneity in forests: interspecific variation in light transmission by canopy trees. *Can. J. For. Res.* **24**: 337–339. doi:10.1139/cjfr-24-2-337.
- Cogbill, C.V. 2000. Vegetation of the presettlement forests of northern New England and New York. *Rhodora*, **102**: 250–276.
- Coomes, D.A., and Grubb, P.J. 2000. Impacts of root competition in forests and woodlands: a theoretical framework and review of experiments. *Ecol. Monogr.* **70**: 171–207.
- Curtis, R.O., and Rushmore, F.M. 1958. Some effects of stand density and deer browsing on reproduction in an Adirondack hardwood stand. *J. For.* **56**: 116–121.
- Doyon, F., Gravel, D., Nolet, P., Bouillon, D., Majeau, L., Messier, C., and Beaudet, M. 2003. L'envahissement par le hêtre dans les érablières de l'Outaouais: phénomène fantôme ou glissement de balancier? Institut québécois d'Aménagement de la Forêt feuillue, Ripon, Québec.
- Duchesne, L., Ouimet, R., and Houle, D. 2002. Basal area growth of sugar maple in relation to acid deposition, stand health, and soil nutrients. *J. Environ. Qual.* **31**: 1676–1683.
- Duchesne, L., Ouimet, R., Moore, J.-D., and Paquin, R. 2005. Changes in structure and composition of maple-beech stands following sugar maple decline in Québec, Canada. *For. Ecol. Manage.* **208**: 223–236. doi:10.1016/j.foreco.2004.12.003.
- Environment Canada. 2007. Climate normals and averages. Environment Canada, Meteorological Service of Canada, Toronto, Ont. Available from www.climate.weatheroffice.ec.gc.ca/climate_normals/results_f.html. [Accessed 15 January 2007.]
- Finzi, A.C., and Canham, C.D. 2000. Sapling growth response to light and nitrogen availability in a southern New England forest. *For. Ecol. Manage.* **131**: 153–165. doi:10.1016/S0378-1127(99)00206-6.
- Forcier, L.K. 1975. Reproductive strategies and the co-occurrence of climax tree species. *Science* (Washington, D.C.), **189**: 808–810. doi:10.1126/science.189.4205.808.
- Foster, D.R. 1992. Land-use history (1730–1990) and vegetation dynamics in central New England, U.S.A. *J. Ecol.* **80**: 753–772. doi:10.2307/2260864.
- Gardner, S.N., and Mangel, M. 1999. Modeling investments in seeds, clonal offspring, and translocation in a clonal plant. *Ecology*, **80**: 1202–1220.
- Hane, E.N., Hamburg, S.P., Barber, A.L., and Plaut, J.A. 2003. Phytotoxicity of American beech leaf leachate to sugar maple seedlings in a greenhouse experiment. *Can. J. For. Res.* **33**: 814–821. doi:10.1139/x03-028.
- Hintze, J.L. 2002. NCSS 2001. User's guide. Vol. 1. Number Cruncher Statistical Systems, Kaysville, Utah.
- Horn, H.S. 1971. The adaptive geometry of trees. Princeton University Press, Princeton, N.J. Monogr. Popul. Biol. 3.
- Horsley, S.B., Long, R.P., Bailey, S.W., Hallett, R.A., and Wargo, P.M. 2002. Health of eastern North American sugar maple forests and factors affecting decline. *North. J. Appl. For.* **19**: 34–44.
- Houston, D.R. 2001. Effect of harvesting regime on beech root sprouts and seedlings in a north-central Maine forest long affected by beech bark disease. USDA For. Serv. Northeast. Res. Stn. Res. Paper NE-717.
- Jackson, M., Pinto, F., Malcolm, J.R., and Wilson, E.R. 2000. A comparison of pre-European settlement (1857) and current (1981–1995) forest composition in central Ontario. *Can. J. For. Res.* **30**: 605–612. doi:10.1139/cjfr-30-4-605.
- Jones, R.H., and Raynal, D.J. 1987. Root sprouting in American beech: production, survival, and the effect of parent tree vigor. *Can. J. For. Res.* **17**: 539–544.
- Kelty, M.J., and Nyland, R.D. 1981. Regenerating Adirondack northern hardwoods by shelterwood cutting and control of deer density. *J. For.* **79**: 22–26.
- Kobe, R.K., Likens, G.E., and Eagar, C. 2002. Tree seedling growth and mortality responses to manipulations of calcium and aluminium in a northern hardwood forest. *Can. J. For. Res.* **32**: 954–966. doi:10.1139/x02-018.
- Krasny, M.E., and DiGregorio, L.M. 2001. Gap dynamics in Alleghany northern hardwood forests in the presence of beech bark disease and gypsy moth disturbance. *For. Ecol. Manage.* **144**: 265–274. doi:10.1016/S0378-1127(00)00391-1.
- Leak, W.B. 1988. Effects of weed species on northern hardwood regeneration in New Hampshire. *North. J. Appl. For.* **5**: 235–237.
- Leak, W.B. 2004. Diameter growth in even- and uneven-aged northern hardwoods in New Hampshire under partial cutting. *North. J. Appl. For.* **21**: 160–163.
- Le Guerrier, C., Marceau, D.J., Bouchard, A., and Brisson, J. 2003. A modelling approach to assess the long-term impact of beech bark disease in northern hardwood forest. *Can. J. For. Res.* **33**: 2416–2425. doi:10.1139/x03-170.
- Long, P.L., Horsley, S.B., and Lilja, P.R. 1997. Impact of forest liming on growth and crown vigor of sugar maple and associated hardwoods. *Can. J. For. Res.* **27**: 1560–1573. doi:10.1139/cjfr-27-10-1560.

- Moore, J.D., Camiré, C., and Ouimet, R. 2000. Effects of liming on the nutrition, vigor, and growth of sugar maple at Lake Clair Watershed, Québec, Canada. *Can. J. For. Res.* **30**: 725–732. doi:10.1139/cjfr-30-5-725.
- Nolet, P., Forget, É., Bouffard, D., and Doyon, F. 2001. Reconstitution historique du dynamisme du paysage forestier du bassin de La Lièvre au cours du 20^{ième} siècle. Institut québécois d'Aménagement de la Forêt feuillue, Ripon, Que.
- Nyland, R.D., Ray, D.G., and Yanai, R.D. 2004. Height development of upper-canopy trees within even-aged Adirondack northern hardwood stands. *North. J. Appl. For.* **21**: 117–122.
- Ouimet, R., and Camiré, C. 1995. Foliar deficiencies of sugar maple stands associated with soil cation imbalances in the Québec Appalachians. *Can. J. Soil Sci.* **75**: 169–175.
- Poulson, T.L., and Platt, W.J. 1996. Replacement patterns of beech and sugar maple in Warren Woods, Michigan. *Ecology*, **77**: 1234–1253. doi:10.2307/2265592.
- Rhoads, A.G., Hamburg, S.P., Fahey, T.J., Siccama, T.G., Hane, E.N., Battles, C., Randall, J., and Wilson, G. 2002. Effects of an intense ice storm on the structure of a northern hardwood forest. *Can. J. For. Res.* **32**: 1763–1775. doi:10.1139/x02-089.
- Ricard, J.-P., Messier, C., Delagrange, S., and Beaudet, M. 2003. Do understory sapling respond to light and below-ground competition? A field experiment in a hardwood forest and a literature review. *Ann. For. Sci.* **60**: 749–756. doi:10.1051/forest:2003069.
- Richards, N.A., and Farnworth, C.E. 1971. Effect of cutting level of regeneration of northern hardwoods protected from deer. *J. For.* **69**: 230–233.
- Robitaille, A., and Saucier, J.-P. 1998. Paysages régionaux du Québec méridional. Publications du Québec, Sainte-Foy, Que.
- Runkle, J.R. 1985. Disturbance regimes in temperate forests. *In* The ecology of natural disturbance and patch dynamics. *Edited by* S.T.A. Pickett and P.S. White. Academic Press, New York. pp. 17–33.
- Sage, R.W., Jr., Porter, W.F., and Underwood, H.B. 2003. Windows of opportunity: white-tailed deer and the dynamics of northern hardwood forests of the northeastern US. *J. Nat. Conserv.* **10**: 213–220. doi:10.1078/1617-1381-00021.
- Saucier, J.-P., Bergeron, J.-F., Grondin, P., and Robitaille, A. 1998. Les régions écologiques du Québec méridional (3^e version): un des éléments du système hiérarchique de classification écologique du territoire mis au point par le ministère des Ressources naturelles du Québec. Ordre des ingénieurs forestiers du Québec, Québec, Que. Aubelle 124. pp. 1–12.
- Seymour, R.S., White, A.S., and deMaynadier, P.G. 2002. Natural disturbance regimes in northeastern North America—evaluating silvicultural systems using natural scales and frequencies. *For. Ecol. Manage.* **155**: 357–367. doi:10.1016/S0378-1127(01)00572-2.
- Siccama, T.S. 1971. Presslement and present forest vegetation in northern Vermont with special reference to Chittenden County. *Am. Midl. Nat.* **85**: 153–172. doi:10.2307/2423919.
- Tremblay, J.P., Huot, J., and Potvin, F. 2007. Density related effects of deer browsing on the regeneration dynamics of boreal forest. *J. Appl. Ecol.* **44**: 552–562. doi:10.1111/j.1365-2664.2007.01290.x.
- Ward, R.T. 1961. Some aspects of regeneration habits of the American beech. *Ecology*, **42**: 828–832. doi:10.2307/1933517.
- Whitney, G.G. 1987. An ecological history of the Great Lakes forest of Michigan. *J. Ecol.* **78**: 443–458.
- Woods, K.D. 1979. Reciprocal replacement and the maintenance of codominance in a beech–maple forest. *Oikos*, **33**: 31–39. doi:10.2307/3544508.